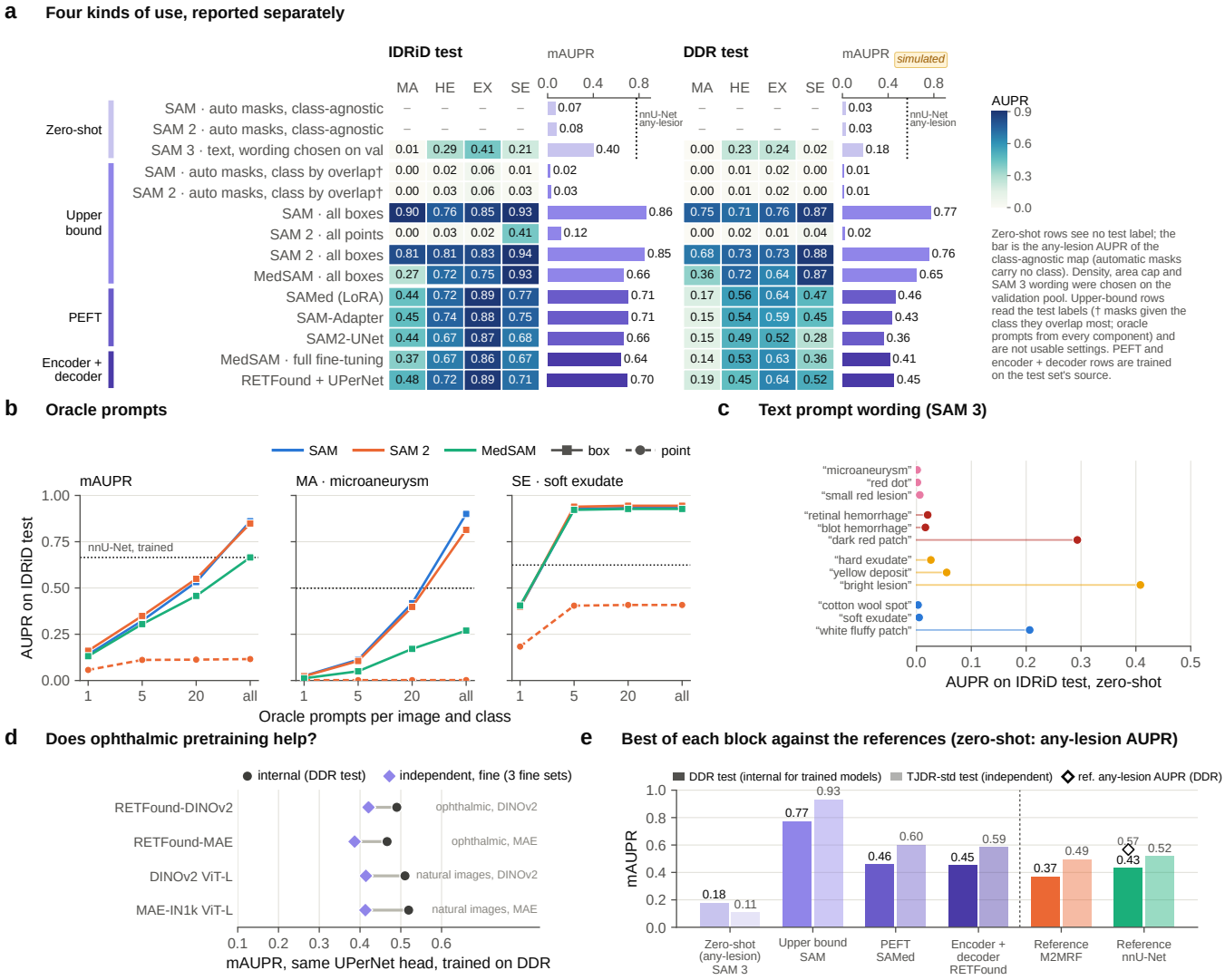

a
Four kinds of use, reported separately
IDRiD test
DDR test
mAUPR
simulated
MA HE EX SE
0.0 0.4 0.8
Zero-shot
Upper bound
PEFT
Encoder + decoder
SAM · auto masks, class-agnostic – – – – 0.07
SAM 2 · auto masks, class-agnostic – – – – 0.08
SAM 3 · text, wording chosen on val 0.01 0.29 0.41 0.21 0.40
SAM · auto masks, class by overlap† 0.00 0.02 0.06 0.01 0.02
SAM 2 · auto masks, class by overlap† 0.00 0.03 0.06 0.03 0.03
SAM · all boxes 0.90 0.76 0.85 0.93 0.86
SAM 2 · all points 0.00 0.03 0.02 0.41 0.12
SAM 2 · all boxes 0.81 0.81 0.83 0.94 0.85
MedSAM · all boxes 0.27 0.72 0.75 0.93 0.66
SAMed (LoRA) 0.44 0.72 0.89 0.77 0.71
SAM-Adapter 0.45 0.74 0.88 0.75 0.71
SAM2-UNet 0.44 0.67 0.87 0.68 0.66
MedSAM · full fine-tuning 0.37 0.67 0.86 0.67 0.64
RETFound + UPerNet 0.48 0.72 0.89 0.71 0.70
SAM · auto masks, class-agnostic – – – – 0.03
SAM 2 · auto masks, class-agnostic – – – – 0.03
SAM 3 · text, wording chosen on val 0.00 0.23 0.24 0.02 0.18
SAM · auto masks, class by overlap† 0.00 0.01 0.02 0.00 0.01
SAM 2 · auto masks, class by overlap† 0.00 0.01 0.02 0.00 0.01
SAM · all boxes 0.75 0.71 0.76 0.87 0.77
SAM 2 · all points 0.00 0.02 0.01 0.04 0.02
SAM 2 · all boxes 0.68 0.73 0.73 0.88 0.76
MedSAM · all boxes 0.36 0.72 0.64 0.87 0.65
SAMed (LoRA) 0.17 0.56 0.64 0.47 0.46
SAM-Adapter 0.15 0.54 0.59 0.45 0.43
SAM2-UNet 0.15 0.49 0.52 0.28 0.36
MedSAM · full fine-tuning 0.14 0.53 0.63 0.36 0.41
RETFound + UPerNet 0.19 0.45 0.64 0.52 0.45
nnU-Net any-lesion
AUPR
0.9 0.6 0.3 0.0
Zero-shot rows see no test label; the bar is the any-lesion AUPR of the class-agnostic map (automatic masks carry no class). Density, area cap and SAM 3 wording were chosen on the validation pool. Upper-bound rows read the test labels († masks given the class they overlap most; oracle prompts from every component) and are not usable settings. PEFT and encoder + decoder rows are trained on the test set's source.
b
Oracle prompts
SAM SAM 2 MedSAM box point
mAUPR
MA · microaneurysm
SE · soft exudate
AUPR on IDRiD test
1.00 0.75 0.50 0.25 0.00
nnU-Net, trained
1 5 20 all
Oracle prompts per image and class
c
Text prompt wording (SAM 3)
“microaneurysm”
“red dot”
“small red lesion”
“retinal hemorrhage”
“blot hemorrhage”
“dark red patch”
“hard exudate”
“yellow deposit”
“bright lesion”
“cotton wool spot”
“soft exudate”
“white fluffy patch”
0.0 0.1 0.2 0.3 0.4 0.5
AUPR on IDRiD test, zero-shot
d
Does ophthalmic pretraining help?
internal (DDR test)
independent, fine (3 fine sets)
RETFound-DINOv2
RETFound-MAE
DINOv2 ViT-L
MAE-IN1k ViT-L
ophthalmic, DINOv2
ophthalmic, MAE
natural images, DINOv2
natural images, MAE
0.1 0.2 0.3 0.4 0.5 0.6
mAUPR, same UPerNet head, trained on DDR
e
Best of each block against the references (zero-shot: any-lesion AUPR)
DDR test (internal for trained models)
TJDR-std test (independent)
ref. any-lesion AUPR (DDR)
mAUPR
1.0 0.8 0.6 0.4 0.2 0.0
0.18 0.11
0.77 0.93
0.46 0.60
0.45 0.59
0.37 0.49
0.43 0.57
0.52
Zero-shot (any-lesion) SAM 3
Upper bound SAM
PEFT SAMed
Encoder + decoder RETFound
Reference M2MRF
Reference nnU-Net